

INTERNSHIP REPORT

**Spacecraft Design Suite Development & Optical
Communication Study**

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1. INTRODUCTION

This report summarizes the work performed during my internship at L&T Precision Engineering and Systems. I developed a Spacecraft Design Suite software application and conducted research on Free Space Optics (FSO) and Quantum Communication payloads for satellite systems.

2. WHAT I DID

2.1 Spacecraft Design Suite (Software Development)

I built a desktop application using Python and Tkinter that integrates multiple spacecraft engineering analyses into a single tool. Previously, these analyses were done in separate spreadsheets. My software consolidates everything with a professional GUI and real-time calculations.

The software has 6 modules accessible through tabs:

- **Power Budget Tab:** Calculates solar array sizing, battery capacity, energy generated vs consumed per orbit, depth of discharge, and recharge time. Users enter cell dimensions, efficiency, orbit parameters, and component loads. Results update automatically.
- **UHF Link Budget Tab:** Analyzes RF communication links between satellite and ground. Shows path loss, carrier-to-noise ratio, link margin, slant range, and Doppler shift. Green/red verdict indicates if the link is closed.
- **Transponder Link Tab:** Models C-Band transponder systems with adaptive modulation. Automatically selects the best modulation scheme based on signal quality and calculates achievable data rate and spectral efficiency.
- **Hardware Power Tab:** Tracks power consumption of every circuit board and component. Shows nominal and worst-case totals with pass/fail against a threshold.
- **Data Budget Tab:** Calculates telemetry rates from all sensors across subsystems (ADCS, OBC, Thermal, EPS, Comms, Propulsion, Mechanisms). Shows total bytes per second and daily data volume.
- **Memory Budget Tab:** Computes storage requirements based on data rate and mission duration. Applies overhead and margin factors, then rounds to practical power-of-2 sizes for procurement.

Key features I implemented:

- **Real-time updates:** Change any input and all tabs recalculate instantly without pressing a button.
- **Unit-aware inputs:** Every field has a dropdown for unit selection (Hz/kHz/MHz, W/mW/kW, km/m). Values convert automatically.

- Editable tables: Users can add, remove, or modify components, hardware items, and sensors dynamically.
- JSON save/load: Complete designs can be saved and loaded for version control and team sharing.
- PDF export: One-click generation of professional design review reports with all results and verdicts.
- System Summary: A dashboard showing all budget pass/fail statuses on one page for quick reviews.

2.2 Reference Materials Used

I used the following spreadsheets and presentations provided by the organization:

- Small_Spacecraft_Power_Budget.xls by Jan A. King for power system calculations.
- Transponder Satellite System Link_Adaptive.xlsx by Jan A. King for transponder modeling.
- data and memory budget.xlsx for hardware power, data rates, and memory sizing.
- MEO KP.xlsx for UHF link budget reference data.
- Presentations on FSO Payloads, Quantum Communication Payloads, and FSO/Quantum combined reference.

2.3 Optical Communication Research

I studied Free Space Optics and Quantum Communication payloads and produced the following deliverables:

- Studied five FSO payload types: Optical Communication Terminal (OCT), Laser Downlink, Optical Crosslink, Hybrid RF-Optical, and Quantum Communication payloads.
- Analyzed Pointing, Acquisition, and Tracking (PAT) systems which maintain microradian-level beam alignment between satellite terminals.
- Documented Quantum Key Distribution protocols: BB84 (single-photon), E91 (entanglement-based), and B92 (two-state simplified).
- Derived complete FSO link budget formulas including slant range, telescope gain, free-space loss, atmospheric extinction, pointing loss, received power, noise, SNR, and link margin.
- Created a conceptual mapping from RF to optical terminology to help engineers transition between domains.
- Documented quantum payload architecture covering ten components from photon generation to classical integration.

3. WHAT I LEARNT

3.1 Technical Skills

- Python software development: I learned to build professional desktop applications using dataclasses, event-driven programming, and modular architecture. I implemented MVC pattern, observer pattern for real-time updates, and strategy pattern for unit conversions.
- GUI development with Tkinter: I mastered custom widgets, scrollable frames, tabbed interfaces, and mouse wheel binding. I built unit-aware input fields with automatic conversion and editable tables with dynamic row insertion.
- Engineering mathematics: I worked with orbital mechanics (Kepler's laws, slant range, precession), link budget analysis (path loss, CNR, G/T, noise figure cascades), power system modeling (solar array sizing, battery DoD, energy balance), and adaptive modulation theory.
- Spreadsheet reverse-engineering: I learned to extract formulas from complex engineering spreadsheets, trace hidden intermediate calculations, and validate software outputs against reference data with tolerance analysis.
- PDF report generation: I used ReportLab to create professional documents with styled tables, colored verdict banners, and formatted sections suitable for design reviews.
- JSON serialization: I implemented custom save/load functionality for dataclass objects with nested lists, enabling design persistence and version control.

3.2 Domain Knowledge

- Satellite power systems: I understand how solar arrays, batteries, regulators, and loads interact in orbit. I can calculate energy balance, depth of discharge, and recharge time for mission planning.
- RF communication links: I understand uplink and downlink budgets, path loss, atmospheric attenuation, Doppler shift, and how combined CNR determines link quality. I can assess whether a link is closed or failed.
- Transponder systems: I understand how satellites amplify and retransmit signals, how intermodulation affects performance, and how adaptive modulation maximizes data rate under varying conditions.
- Free Space Optics: I understand how laser communication works in space, why narrow beams provide security and high bandwidth, and how PAT systems maintain alignment. I know the trade-offs between optical and RF systems.
- Quantum communication: I understand QKD protocols, entangled photons, the no-cloning theorem, and how quantum mechanics enables fundamentally unbreakable encryption. I can describe payload components from photon source to detector.

- Data and memory budgeting: I understand how to calculate telemetry rates from sensor configurations, apply overhead and margin factors, and size storage for mission requirements.

3.3 Professional Skills

- Technical documentation: I wrote comprehensive README files, user guides, and this report. I learned to explain complex engineering concepts clearly for different audiences.
- Validation methodology: I developed systematic testing approaches comparing software outputs against reference spreadsheets cell-by-cell, ensuring zero deviation before declaring a module complete.
- Research skills: I reviewed technical presentations, extracted key concepts, and produced independent documentation including formula derivations and architecture descriptions.
- Problem-solving: I encountered challenges like Tkinter scrolling limitations, unit conversion complexity, and real-time update lag. I solved each through research, experimentation, and iterative refinement.
- Project management: I delivered multiple modules on schedule, managed version history (gui_v1 through gui_v24), and archived stable releases as executable files.

4. CHALLENGES AND HOW I SOLVED IT

- Spreadsheet formulas were hidden and nested: I traced calculations cell-by-cell, tested boundary conditions, and validated every output against known reference values.
- Units were inconsistent across spreadsheets: I built a centralized unit family system with automatic conversion so users never make unit errors.
- Tkinter has no native scrollable frames: I created a custom ScrollableFrame class using Canvas and Scrollbar that works across all tabs.
- Real-time updates caused UI lag: I added an updating flag to prevent recursive calculations and optimized the calculation order.
- RF and optical engineers use different terminology: I created an explicit mapping table so both domains can understand each other.

5. CONCLUSION

This internship provided hands-on experience in both software engineering and aerospace systems. I delivered a production-ready spacecraft design tool that replaces multiple spreadsheets with a unified application, and produced technical documentation on optical and quantum communication systems. The skills I gained in Python development, satellite engineering, and technical communication will support my future career in aerospace.